

Industrial intrusion detection based on the behavior of rotating machine

Mohammad Safari^a, Elham Parvinnia^{a,*}, Alireza Keshavarz Haddad^b

^a Department of Computer Engineering, Shiraz branch, Islamic Azad University, Shiraz, Iran

^b School of Electrical and Computer Engineering, Shiraz University, Shiraz, Iran

ARTICLE INFO

Article history:

Received 14 July 2020

Revised 7 January 2021

Accepted 5 February 2021

Available online 21 March 2021

Keywords:

Cyber security

Industrial control system (ICS)

Cyber physical system (CPS)

Behavioral intrusion detection system

Rotating machine

ABSTRACT

In this study, a new industrial intrusion detection method is introduced for the control system of rotating machines as critical assets in many industries. Data tampering is a major attack on the control systems which disrupts the functionality of the asset. Hence, our objective is to detect data manipulations in the system. We use the behavior of the rotating machine to propose new industrial intrusion detection for the control system of the rotating machine by machine learning techniques. The behavior is elicited by the data of sensors under all the conditions of the rotating machine operation. In this work, the nonlinear regression, novelty detection, outlier detection, and classification approaches are implemented to create behavioral model. On each implementation, online data are compared with the real data of behavior prediction model during the operation of the rotating machine to detect any abnormality. According to our experimental results, the accuracy of the behavioral models created by the One-classSVM novelty detection, k-Nearest Neighbor (kNN) outlier detection, decision tree classifier, k-Neighbors classifier, random forest classifier, and AdaBoost classifier is obtained as 0.98, 0.994, 0.999, 0.999, 0.999, and 0.999, respectively. The results indicate that the proposed industrial intrusion detection method is able to detect the data tampering attacks on the control system of the rotating machines very accurately.

© 2021 Published by Elsevier B.V.

1. Introduction

Physical world is controlled by the Industrial Control System (ICS); therefore, some significant risks must be considered, including the "safety of human lives, serious damage to the environment, and production losses" [1–3]. Consequently, the ICS is considered as a central important part of critical infrastructures. Security protections must be performed to maintain the integrity and safety of the physical system during normal operations and cyber-attacks [4]. ICS manufacturers utilize the advanced information technology to increase the corporate connectivity and remote access capabilities, minimize the operational overheads, and also achieve the optimum resource utilization and market globalization [5]. Therefore, ICSs have become more similar to the Information Technology (IT) systems. Unfortunately, this has caused significantly less security for ICS than the predecessor systems. According to the Kaspersky report in the first half of 2019, 41.6% of ICS computers in the energy sector were infected by the malicious activities [6]. Thus, there is an urgent need to secure these systems. Strategies,

including those related to the people, processes, and technologies are needed to enhance the security of ICS comprehensively.

Security solutions that have been designed to be used for ICSs cannot be compared to the IT systems [7]. Consequently, any solution must be used without interfering with the system, and for instance, the anti-virus software applied in the IT systems cannot be easily installed and used in the ICS [4]. Unfortunately, cyber security issue has not been considered sufficiently in most of the ICSs [3,8]. In many cases, designers of the ICSs believe that internet disconnection is the only solution for cyber security.

In the recent attacks on the control system, data tampering has become one of the main goals of an attacker, especially in rotating machines as a critical asset in many industries. There are some critical input-output signals in the rotating machines control system, which can disrupt the performance if they are tampered by a hacker or malware. Also, this can create significant harm on the rotating machine or industrial plant. For instance, Stuxnet would be classified as a Cyber-Physical System (CPS) malware developed to harm centrifuge machines [8,9].

Traditionally, Network-based Intrusion Detection Systems (NIDSs) are used to deal with the network intrusion using signatures of the attacks or traffic behavior. Incidentally, the attacker can exploit regular traffic (without any signature or change on traffic pattern) by varying some critical input-outputs; therefore,

* Corresponding author.

E-mail addresses: safari.md@gmail.com (M. Safari), parvinnia@iaushiraz.ac.ir (E. Parvinnia), keshavarz@shirazu.ac.ir (A.K. Haddad).

it seems impossible to detect the data tampering attack using a conventional IDSs.

For addressing this concern, some efforts have been made to introduce new intrusion/threat detection methods for Cyber Physical System (CPSs) according to the process-oriented behavior-based intrusion detection system [10,11]. These efforts are presented with more details in Section 2.

This paper introduces a method for industrial intrusion detection systems based on the behavior of the rotating machines. Most papers have used some predefined data-sets or data from testbed to create the model. In most of the rotating machines the physical and process features are not exactly same even in machines with same model and brand. Therefore, there is a concern to use same dataset or predefine dataset which was created from real data of test bed to implement intrusion detection system specially to detect data tampering attack for all rotating machines. In this study this concern is addressed. We are using the real data from operational machine (not test bed) to implement the IDS for data tampering attack. Unlike the others papers the data of this work have been extracted from the real rotating machine under all the conditions of normal operation. The proposed method extracts the data from instrumentation sensors and devices directly. To do this, the physical signals are taken out by installing the signal duplicator in the control system.

Behavioral models of the normal rotating machine are created by the nonlinear regression, novelty detection, outlier detection, and classification algorithm. Then, the online data are compared with the data of the prediction model of behavior during the operation of the rotating machine to detect any abnormality. Next, the nonlinear regression, novelty detection, supervised and unsupervised outlier detection, and classification algorithm are used to prove which one has efficient results in creating of the behavioral model of the rotating machine.

The experimental results indicate that the proposed method can detect the tampering data attacks on the rotating machine control systems very accurately. nonlinear regression and novelty detection employ the original dataset only consisting of regular records, which is considered as a strength and advantage for these algorithms. The nonlinear regression model implemented by the four-layer fully-connected (dense) neural network has the Mean Absolute Error (MAE) of 1.5%. The accuracy of behavioral models created by the One-Class Support Vector Machine (SVM) novelty detection, kNN outlier detection, decision tree classifier, and k-Neighbors classifier, random forest classifier, and AdaBoost classifier are equal to 0.98, 0.994, 0.999, 0.999, 0.999, and 0.999, respectively. The other evaluation metrics show that classification and outlier detection are more efficient than the novelty detection and nonlinear regression. Test time values demonstrate that all the methods work in a timely manner compared to the control system scan time.

The contributions of this study are as follows:

- Proposing a new method to detect any harmful data manipulation on the rotating machine control system based on the behavior of the rotating machine in the normal operational mode.
- Utilizing the real process data in the normal operational mode to propose the industrial intrusion detection on the control system of the rotating machines instead of simulation or making use of pre-existing datasets.
- Trajectory evolution to identify the critical signals of the rotating machines.
- We describe some appropriate features to create a data set that is suitable for use in machine learning to generate an intrusion detection system in the rotating machines.
- Nonlinear regression, novelty detection, outlier detection and classification methods have been used together in this study. different algorithms of these methods have been applied to

produce a data manipulation detection system and compared the results of the methods.

- Based on the experimental results obtained in this work on a number of algorithms and based on the comparison of different metrics and score, very accurate detection method are presented to detect data change attacks, at least for rotating machines.
- Implementation of the IDS is control system independent and conveniently without having in-depth knowledge of the industrial network.
- Suggesting to extract the data directly from the sensor and actuator through hardware signals using a signal duplicator to guarantee the validation of received data in the IDS.

The other parts of this paper are arranged as follows: In the next Section 2, the related works are discussed. Section 3 presents the initial definition regarding the control system and case study of the rotating machine. The proposed method and experimental results are provided in Section 4. Section 5 summarizes the study findings, and provides the conclusions.

2. Related work

2.1. Attacking techniques on ICSs

Security vulnerabilities and the importance of control systems have made them an attractive target for hackers. Control systems are also vulnerable to the general attack methods due to their security vulnerabilities, for instance, buffer overflow, code injection, Structured Query Language (SQL) injection, and data manipulation [8].

In data manipulation attacks, control commands are compromised by deception attacks (false data injection) [1,12]. This kind of attack can be performed by transmitting fake information from a sensor or other components to the controller [13].

Li et al. [14] investigated the design problem of data-injection attacks in the CPSs. Kull- back-Leibler Divergence (KLD) was used to develop a new method for the data-injection attack.

Magdiedet et al., studied on the model of "Denial-of-Service (DoS), deception, and replay attacks" in the CPS [15]. They investigated the approaches of filtering the CPS and controlling it with respect to these attacks.

Security issues of the Programmable Logic Controller (PLC) firmware was studied in several works, including the study by Milinkovic et al. [16]. Basnight et al. used the reserve engineering [17] to recognize the firmware weaknesses. Schuett et al. [18] appraised the firmware to find the vulnerabilities and backdoors in the firmware of PLC and its peripheral.

The attacks on the ICS have become more complicated in the past decade, including the rootkits and advanced persistent threats. Rootkits are designed to be unrecognizable and are usually used as part of a multi-stage attack [8].

Stuxnet was developed as the first rootkit for the ICS, in which the rootkit was applied to manipulate the programming of the Iranian centrifuges PLCs [9]. It has been classified as a CPS malware developed to harm a centrifuge. It was used to increase the internal pressure and altering the rotor speed of the centrifuge by changing the frequency of the electrical current to the drives that both were hidden from the operator's eyes. It means that the physical effects of the Stuxnet were hidden as it exploited the targeted system [8].

2.2. Intrusion detection system (IDS) for ICSs

The primary step to overcome the intrusion is discovering it after its occurrence. Some conventional IT intrusion detection methods have been tailor-made for ICS, although ICS system parts are

much distinct from those of IT components. Colbert Edvard et al. [8] reviewed traditional types of IDS, and their usage in control systems. Host-Based Intrusion Detection Systems (HIDS) are rarely used in the ICS, as the ICS components have never been developed to accommodate the additional software. Network-based IDSs extract useful data from raw unformatted network packet data directly from the network. Two methods have been used for the NIDSs, including i) Signature-based NIDS that compares the network traffic with specific bit signatures to detect known malicious code. "Signature-based detection methods are also considered as misuse-based detection or knowledge-based detection methods". (ii) Non-signature-based NIDS methods analyze the network traffic with respect to the behavioral patterns. They are sometimes considered as behavior-based or model-based detection methods. Non-signature-based NIDS is able to detect unknown malicious activities. Complex implementation and higher false positive rate are the drawbacks of this method. Signature-based IDS can be used for ICS networks by the specified rules for the network between workstations and PLCs. Signature-based IDS is not sufficient for detection, as it may ignore a new attack; therefore, the researchers have developed non-signature-based IDS to promote the security simultaneously [8].

Multi-agent IDS architecture has been designed based on a biologically leaded learning mode for decentralized intrusion detection. The Ant Colony Clustering Model (ACCM) has been improved for near-optimal clustering heuristically. This method has been proposed for industrial networks, although the Knowledge Discovery in Databases (KDD)-Cup99 dataset has been used for the IDS benchmark [19].

Cheung et al. developed a model-based IDS that characterizes the expected behavior of the system [20]. This method was designed according to the fact that "the control networks have static topologies, regular traffic patterns, and a limited number of applications and protocols running on them". The prototype of these models was applied for monitoring the Modbus-TCP networks.

Oman and Phillips developed a customized intrusion detection method for IDS of an ICS [21]. As this method was designed based on the full ICS device functionality; therefore, the system raised the alerts accurately for unauthorized access and changes in the device settings.

An IDS has been developed to detect the complex attacks on the Supervisory Control and Data Acquisition (SCADA) systems using the state evolution and Fovino et al. designed the rule language to express the system's critical states [22]. The prototype of the proposed IDS was developed to monitor the systems using the Modbus and Distributed Network Protocol 3 (DNP3) communication protocols.

Yang et al. proposed the industrial IDS for SCADA consists of the signature-based and model-based detection methods [23]. This method introduced a rule-based IDS for industrial networks using Deep Packet Inspection (DPI). The abnormal condition of the ICS can be discovered by distinguishing whether the measurement and control data are in a valid range [24]. It is impossible to detect the data tampering by the NIDSs as these attacks mostly do not manipulate the traffic features [25]. Therefore, the researchers have attempted to design the Process-Oriented Intrusion Detection system.

Ding et al. [12] presented an overview of the recent advances in the security control and attack detection of an ICS. They investigated the "DoS attacks, replay attacks, and deception attacks" from different points of view.

2.3. Process-oriented behavior-based ids approach

An IDS can be developed based on a process behavior for important assets in the industry to enhance the security of CPSs for

overcoming the threats that manipulate the data within an acceptable range. Many works have been done related to the industrial intrusion detection system in the last decade in the field of industrial network IDS and behavioral model of the control system [12,26,27]. Specifically, Urbina et al. [28] presented the theoretical and practical contributions to the trajectory evolution of physics-based attack detection in the control systems. They illustrated a comprehensive view of the related researches. The state of the art of the reviewed techniques focusing on the tampering of the control system output is presented in the following with more details.

Zhanwei and Zenghui [29] introduced "an abnormal detection method based on a behavioral model". Their model included the elicited behavioral data sequences and they constructed a normal behavioral model of the control system and site, which is controlled. Control system and site behavior output sequences were portended according to the normal behavioral models. Tampering and intrusion attacks were discovered by comparing the online exploited behavioral sequences with the predicted behavioral output sequences. The data were extracted from the Modbus-TCP network. A chemical mixing reaction platform was used as testbed.

Sicardet et al. [1] suggested a method to detect the malicious orders in the discrete system. In their proposed method, four types of attacks "(direct, sequential, temporal, and over-soliciting)" were studied. The "S.A.F.E. (Security Approach based on Filter Execution)" was used to detect the malicious orders transferring from the control system to the physical system. The acceptable efficiency of the filter-based approach to detect the cyber-attacks in lower levels of ICS was also proved. The results were simulated in the industrial platform as well. The improvements of their method included the prevention of outputs before occurrence of any critical changes in the system condition and detection of sequential attacks using the combinational or temporal constraints. Some points were also proposed for the future works to increase the detection rate. The evolution of the system between two stable states, lack of security analysis on design of the filter, lack of enough data to use the machine learning approach during the development of the models, and protection between the control and supervision parts were the main requirements of this method that need to be improved. They focused only on discrete-event systems.

Zhang et al. [30] proposed a multiple-layer detection system that followed the concept of defense-in-depth. The proposed multiple-layer detection system exploited the network, system, and process data to provide the multiple layers of cyber detection for an ICS to prevent the damages to the physical system. AAKR (Auto-Associative Kernel Regression) was applied by the physical process data to detect the undetected attacks and threats by the network and system data detection especially "command tampering and false data injection attacks". The data used for constructing the detection system were extracted from a real-time ICS test bed. The detection results obtained using the data extracted from the mentioned explicit test bed of the proposed method had acceptable detection accuracy.

Han et al. [2] discussed the methods applying the intrusion detection mechanism to the CPS. They introduced an intrusion detection mechanism, which can be integrated into the CPS. A design framework was offered by inspecting the unique properties and requirements for reliability and security in the CPS.

Kravchik et al. [31] introduced an IDS on ICS based on unsupervised deep neural networks. For this purpose, they used the convolutional neural networks and the Secure Water Treatment test bed (SWaT) dataset. Their proposed method improved the detection rate compared to the previous works on this dataset. It confirmed the usage of 1D Convolutional Neural Networks (CNN) for detecting the cyber-attacks in the ICS data with low false positives.

Li et al. [32] employed the physical process data for the cyber security in the nuclear power plants. Statistical analysis algorithms

were applied to the physical data to detect the anomaly caused by the cyber-attacks.

Adepu et al. proposed a distributed detection method [33] to “detect the Single-Stage Multi- Point (SSMP) attacks on a CPS”. Basically, the goal of this kind of attack is endangering the instruments sensors or actuators at any level of a CPS. The proposed method focused on the malicious manipulation of sensor data, or the condition of actuators by a man-in-the-middle attack. The method was designed based on the physical invariants extracted for each stage of the CPS. The attack detection was appraised by an operational water treatment test bed.

Settanni et al. [3] suggested an anomaly detection technique along with embedded self- protection for the Cyber-Physical Production Systems (CPPS). They illustrated an anomaly detection mechanism that can be utilized to protect the CPPS against the advanced threats by self-adaptation demonstration (automatic configuration changes). “Monitor-Analyze-Plan- Execute over a shared Knowledge (MAPE-K)” reference model was used for the self-adapting mechanism in this work.

Adepu et al. [11] experimentally investigated the effectiveness of attack detection using the process invariants. The operational water treatment system was used to implement this research. They showed the appropriate values of the selected parameters as focal points in this approach that depend on the system dynamics.

Wu et al. studied the use of “machine learning on the physical data for detecting the cyber- physical attacks” [25]. 3D printing and Computer Numerical Control (CNC) milling machine was utilized to implement the “machine learning methods in the physical data” for detecting the cyber-physical attacks. In the 3D printing, the vision was utilized as a data source for machine learning. They proved that the anomaly detection method has the accuracy of 96.1% in recognizing a malicious defect in the printing process. In the “CNC milling process”, an acoustic signal was chosen for the machine learning process as a source of physical data. They illustrated that the Random Forest Algorithm had the highest average accuracy of 91.1% in detecting a malicious defect in the CNC milling process.

Wu et al. proposed an “intrusion detection system for the cyber-manufacturing system” [34]. In this method, merging the value of “physical data from the process and systems with cyber data from network” was suggested along with hosting to develop the intrusion detection with low false alarms.

Clotet et al. [10] proposed the design of “an anomaly-based detection system for real-time detection of abnormal behaviors” caused by the attacks to getting a goal to “the industrial process level of critical infrastructures”. The detection method was designed according to the negative selection algorithm. The proposed “PR-IDS” exploited the input data extracted from the measurements extracted at various points of the process and carrying out the real-time analysis. They claim the accuracy of 85% for the system can be achieved by a public dataset from a water treatment plant laboratory.

Huang et al. [35] presented a new integral sliding-mode function method for a class of actuator attacks. The effectiveness of the proposed control strategy was proved by a numerical example. The detector was designed to track the corresponding process variables and prediction models as the basis for assessing the future activity by Hadziosmanovic et al. [36]. The evaluation of this detector was appraised by the traffic data recorded at two operational water treatment plants.

In addition, some papers have applied physical based models of normal operation to detect attacks. An anomaly-based process-level intrusion detection system presented by Aoudi et al. [37]. In this work, ICS process activity is monitored in real time to recognize the system operation is normal or anomalous. Stealthy-attack detection was used to detect structural changes in the process be-

havior. The proposed method was tested by data from the “Secure Water Treatment (SWaT) plant”. Urbina et al. [38] presented stealthy attacks detection method base on physics- based attack detection. They focused at the physics of the process under control not devices. Cardenas et al. [39] proposed how to use of “automatic response mechanisms” based on the normal state of the system. They showed that “incorporating a physical model of the system” can detect computer attacks that manipulate the behavior of the physical process. The main motivation of this study is how to develop intrusion detection systems based on process data in major industrial assets. The present paper focuses on a type of widely used rotating machine (compressor) as one of the important assets, elaborating that how one can use proper sensor data and machine learning techniques to detect abnormal/dangerous state on the compressor. We take a new approach on applying process data for intrusion detection, although process data has already been used in other work.

2.4. Contribution of this study

Most of existing works have used the predefined data set or data from test bed to create the model. We argue that in rotating machines (Compressor, Gas Turbine,) the physical process features are not unique for each machine. In the rotating machines the physical and process features are not exactly same even in machines with same model and brand. Therefore, there is a concern to use same dataset or predefine dataset which was created from real data of test bed to implement intrusion detection system specially to detect data tampering attack for all rotating machines. We are using the real data from operational machine to implement the IDS for data tampering attack. Therefore applying real process data extracted from an operational plant to develop a new intrusion detection method for such a critical asset (rotating machine) is the main contribution of the present research. In this work, we explain how to use raw process data and determine the points that can be exploited by an attacker, as well as how to select features for use in an intrusion detection system based on the behavior of a rotating machine. The paper studies on a type of widely used rotating machine (compressor), elaborating that how one can use proper sensor data and machine learning techniques to detect abnormal/dangerous state on the compressor. Any control loop, especially critical conditions must be tuned in commissioning phases of an industrial plant; therefore, utilizing the long-term process data, including all the operational conditions can improve the trajectory of intrusion detection instead of using the design data. The new training dataset can be applied to the model conveniently; consequently, alteration of the process can be managed by this method. Also, the data are directly extracted from the sensor and actuator through hardware signals using a signal duplicator. One can make sure to achieve the actual data from the sensor or actuator by the IDS.

Most of ICSs in the operational industries are outdated; hence, it is not possible to implement the new strategy of cyber security on them. The proposed method looks at the control system as a black-box and detects any abnormality due to cyberattack without requiring deep knowledge of the rotating machine. Thus, the method is simple and easy to implement in comparison with the behavioral model of the control system which focus on the network traffic or all system parameters. We describe appropriate features to create a data set that is suitable for use in machine learning to generate an intrusion detection system in the rotating machines. Nonlinear regression, novelty detection, outlier detection and classification methods are used together in this study. Various algorithms are applied to produce a data manipulation detection system and the results of the methods are compared. Based on the experimental results obtained in this work on a number

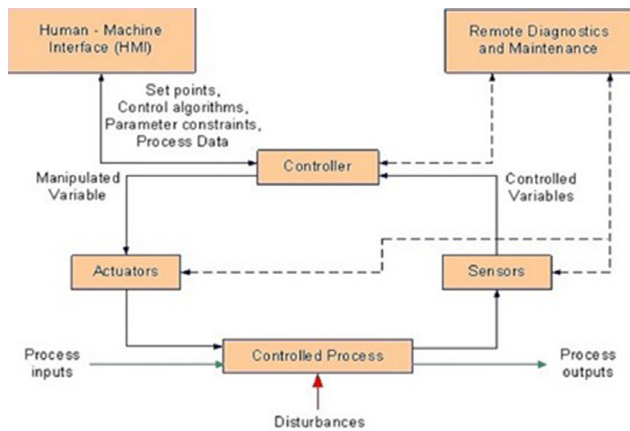


Fig. 1. The basic operation of the controller.

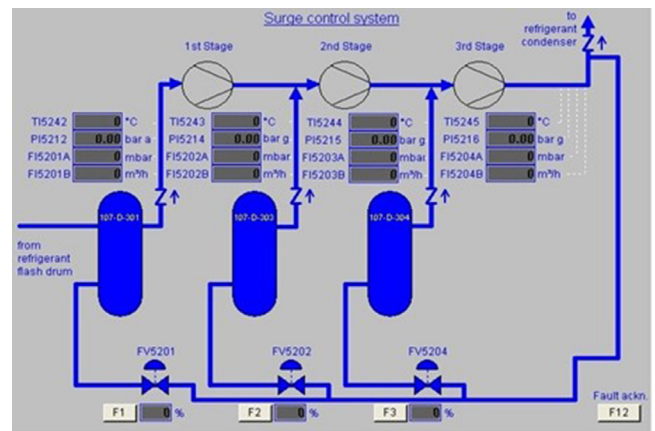


Fig. 2. Surge control scheme of the compressor.

of algorithms and comparison of different metrics and score, very accurate methods for data tampering are presented, at least for rotating machines.

3. Preliminary definitions

3.1. Industrial control system (ICS)

An ICS is working as automated controlling equipment in the manufacturing and chemical plants and many other industries. ICS consists of industrials PCs, digitalized controllers, sensors known as primary elements and electro mechanical actuators functioning as final elements. A robust network serves among controllers and PCs as a real time network to give ability of data exchange between any network elements [8].

There are four types of ICS controllers: "Programmable Logic Controller (PLC)", "Process Control System (PCS)", "Safety Instrumented System (SIS)", and "Distributed Control System (DCS)". The PLC is frequently used in SCADA and DCS as a component. The PLCs are also exploited as the primary controller in the smaller control system configurations to supply the operational control on electromechanical equipment such as rotating machines (compressor) and power plants (gas turbine) [7].

The "Engineering Work Station (EWS) and Human Machine Interface (HMI)" are typically considered as a desktop computer running a standard operating system such as Microsoft Windows or Linux. The EWS has been used by the engineers to create and modify the controller logic. The HMI is used by the plant operator to supply the situational awareness of the plant such as process values (temperature, flow, and tank levels), alarms, and data trends [8].

A control loop is used to regulate some processes. A physical property of the process is measured by the sensor and then, it is transmitted as the controlled variable to the controller. The controller executes the logics and produces the corresponding regulated variables. Regulated variables are transmitted to the actuator as a controller command. Actuators such as control valves, gear-boxes, breakers, switches, and motors are utilized to adjust the controlled process based on the commands from the controller. Fig. 1 shows the basic mentioned operation and the relationship between the components [7].

Nowadays, electronic controllers are used on the rotating machine. When a digital controller is utilized, its scan time (the time required to execute a logic cycle) should be 0.3 s or less [40]. Therefore, a PLC or PCS can be used as a rotating machine control system.

Some of the above-mentioned terms have a similar meaning in the scope of this paper and can be used interchangeably.

If the attacker succeeds in gaining access to any part of the ICS, the tampering data attack will take. The aggressor can manipulate the output signals from the controller to any actuator (control valve, relay, and electric motor) to damage the compressor. If the changes are in normal range of signals, the manipulation detection is more complicated.

3.2. Case study of rotating machine

In this study, a three-stage centrifugal compressor (Type10 MH 6A developed by the Demag Delaval Turbo Machinery Co., USA) was used as an example of the rotating machine for the gas pressure boosting. Compressors are valuable rotating machines used to increase the gas pressure. Compressors are usually powered by an electric motor connected to a compressor via a gearbox. The compressor studied in this paper was a three-stage centrifugal compressor used to increase the pressure of propane gas in a closed loop.

One of the main goals of controlling a compressor is preventing the compressor from working in surge mode. The surge protection system of this compressor is consisted of the three surge control valves, as shown in Fig. 2, each of which was installed in a step, as well as the surge protection control system. The gathered data were measured using the precision instruments and sensors and were transmitted to the control system.

The surge control system is configured based on the performance curve of the compressor calculated by the manufacturers for each compressor [40]. Fig. 3 demonstrates this curve as an example. The main purpose is controlling the compressor near the efficiency line and preventing it from entering the surge area. This means that the more the compressor works near the efficiency line, the higher the efficiency. There is also a risk of crossing the efficiency line and entering the surge area; therefore, it must not exceed the surge area while maintaining efficiency. a surge occurs when the flow through a compressor is less than a minimum at a certain differential pressure. If the compressor detects the entry to the surge area, it opens the anti-surge control valves for each stage to come out from the surge area by compensating the minimum flow requirement.

If these valves are opened gradually, it indicates that the compressor has not entered the surge section and it is close to the performance curve, as a result, the controller gradually increases the opening of the anti-surge valves as needed to prevent entry into the surge section. Therefore, if the compressor is operating normally, the measurement of the data will be expected to follow a

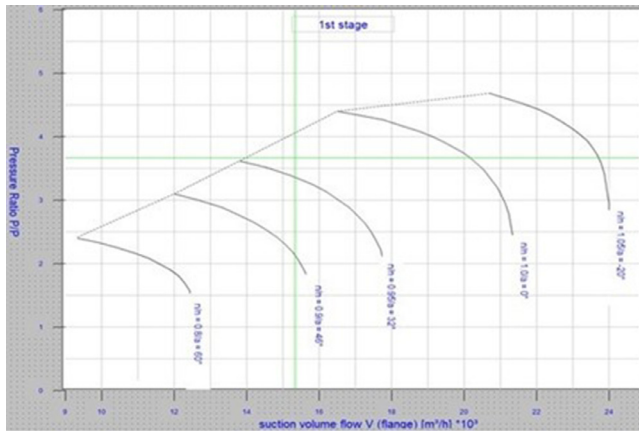


Fig. 3. Performance curve and surge area.

certain pattern. If an aggressor intends to harm the compressor in any way, the attacker should try to extract the control data from that particular pattern. For instance, a cyber-physical attack scenario can be as follows: manipulating the control data and hiding the actual data when the compressor is close to the surge area and the control valves should be opened to the extent determined by the control system. If the penetration of this scenario is successful, it will cause physical damage to the compressor.

4. Proposed method

The proposed industrial intrusion detection system is based on the behavior of some input–output values of the rotating machine. The behavioral model of the rotating machine is created according to the critical signals of the control system, which can be manipulated by the aggressor to damage the rotating machine. These signals should be validated in the intrusion detection system; therefore, the signal duplicator is used to transfer the critical signal to intrusion detection safely. The selection of critical signals is among the main concerns in this method, which is discussed in Section 4.1 in details. The behavioral models of the rotating machine were implemented by the nonlinear regression, novelty detection, outlier detection, and classification algorithm to assess in real system in which one provides higher performance. In this study, the following steps were implemented to develop an effective model regarding the intrusion detection system based on the behavior of the rotating machine.

1. Investigation to select the critical signal of the rotating machine, which can be tampered by the aggressor to blemish the machine or interrupt the normal operation.
2. Presenting the data related to the selected critical signal in terms of operation, safety, and physical effect.
3. Obtaining all the condition of the rotating machines in normal state.
4. Extracting the data in all the conditions of the rotating machine.
5. Creating the dataset according to the above steps.
6. Implementing the methods based on the nonlinear regression, novelty detection, outlier detection, and classification algorithm.
7. Selecting the best approach according to the evaluation metrics, training, and test time. The first five steps are discussed in Section 4.1.

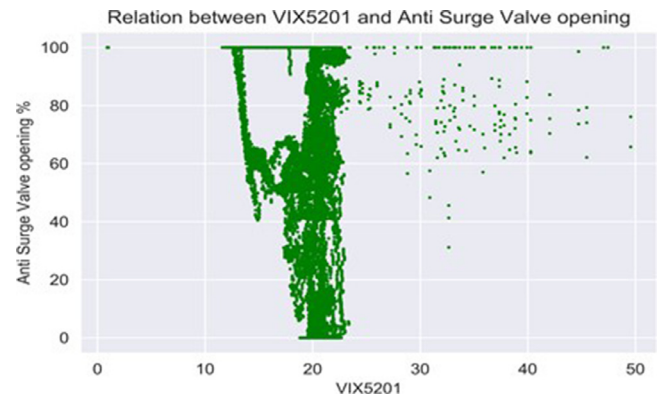


Fig. 4. Drawing the paired control charts regarding the relation between the valve opening and vibration point.

4.1. Preparation of the proposed dataset

In the studied compressor, all the measured data of the compressor status on the control system were sampled and stored during 2014–2016 at every 500 ms in all the available conditions such as startup, shutdown, loading, unloading, and normal operation.

Since, the control system of this compressor was of Siemens PCS7 version 8 (417H) type; therefore, the storage and sampling were done using the Siemens monitoring software called WINCC to store the data of precision instruments. The number of selected data records with the available representative for all the conditions is equal to 200,000. The initial number of features stored in each record is equal to 90. The control system controls and manages the performance of this compressor by the three commands for anti-surge control valves and two commands for gearbox speed and the opening rate of the compressor inlet blade. Aggressors attempt to manipulate these five critical outputs.

In this study, the first-stage anti-surge control valve was selected as the target sample and subsequent studies were performed on it. Obviously, the same process can be considered for other outputs.

In the next step, it was necessary to choose from all the features having a significant relationship with the status of the first-stage anti-surge control valve. The number of process features related to the first-stage anti-surge control valve can be selected based on the manufacturer's documents; hence, four features were selected. The opinions of the rotating machine experts were used as well as drawing the paired control charts to select indirect and non-process characteristics. The experts opinion and paired control charts confirmed the relationship between the opening of the anti-surge valve and the value of the vibration sensor on the compressor. Fig. 4 illustrates the relationship between the opening of the anti-surge control valve and the amount of one of the vibration points. A scan can be seen, when the vibration rate is more than 25, the control valve must be opened by more than 50%.

Therefore, the data of six vibration points were added to the previous set of features, and finally, the dataset had 11 features. Fig. 5 depicts the location of vibration points in the compressor.

The advanced feature selection methods can be employed for a more complicated problem.

Table 1 presents the final dataset with the selected feature.

All of the above records are related to the normal mode operation without any changes in the data caused by the intrusion. Therefore, all these records can be considered as a normal class. This data set, which has been created so far, will not be efficient enough to be used for the methods requiring both normal and abnormal classes in the dataset. Drawing the paired chart of anti-surge control valves and other features was done to create the

Table 1
Dataset with the selected feature in the present study.

Row	Tag	Description	Range	Eng Unit
1	FIC520LFinaloutput	Output from controller to Valve1	0–100	Percent
2	PT5212_V	First Stage Suction pressure	–11	Bar
3	PT5214_V	First Stage Discharge pressure	0–10	Bar
4	FT520LV	First Stage Flow of Gas	0–100	Mbar
5	TT5242_V	First Stage Gas temperature	–200	Deg C
6	VIX5201	Compressor vibration	0–150	Micro meter
7	VIY5201	Compressor vibration	0–150	Micro meter
8	VIX5204	Compressor vibration	0–150	Micro meter
9	VIY5204	Compressor vibration	0–150	Micro meter
10	ZI5202A	Compressor vibration	–2	mm
11	ZI5202B	Compressor vibration	–2	mm

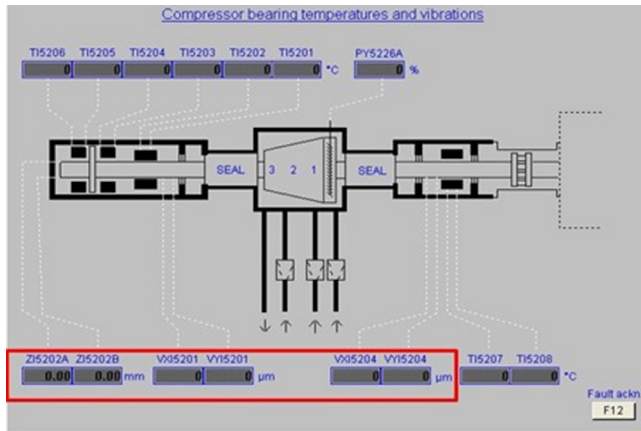


Fig. 5. The location of vibration points in the compressor.

manipulated records. We use paired chart interpretation to create an unusual records. This should be consistent with the tampered records made by each attacker for sabotaging the machine. Our attack model includes any data manipulation in the control system. This can be a change in sensor data or controller output to a physical process (Actuator). Since any small change in the relationship between the opening of the anti-surge valve and other related features is considered an abnormal history, we expect to detect stealth attacks as well. We consider the transmission of sensor signals using a hard wire (4–20 mA) to the control system (this is the real situation in most industries) so attacking the sensor itself is beyond the scope of this study but manipulating the amount of the sensor in the control system is included in our attack model. For instance, as shown in Fig. 4, if the vibration is more than 25, the valve must be opened by more than 50%; therefore, the attacker manipulates the opening of the anti-surge valve by less than 50%. As demonstrated in Fig. 6, when the output/input pressure ratio is more than 2.7, the anti-surge valve opening is equal to 100%; hence, the intruder also changes this value if he intends to sabotage. It should be noted that the two mentioned modes and other ones used to create the manipulated record are mechanically problematic for the rotating machine, which caused the physical damage. Thus, the minimum changes were made in the manipulated records compared to a required change by the aggressor to promote the efficiency of our model in the real cases. In fact, paired diagrams show the status of these parameters in normal operation. Therefore, strictly, any change in circumstances can be considered as a cyber attack effect.

The manipulated records were added to the dataset using the stated method, and finally, the data set, including both classes was created with 220,000 records. This dataset was prepared for this particular type of the rotating machine and a sensitive output of

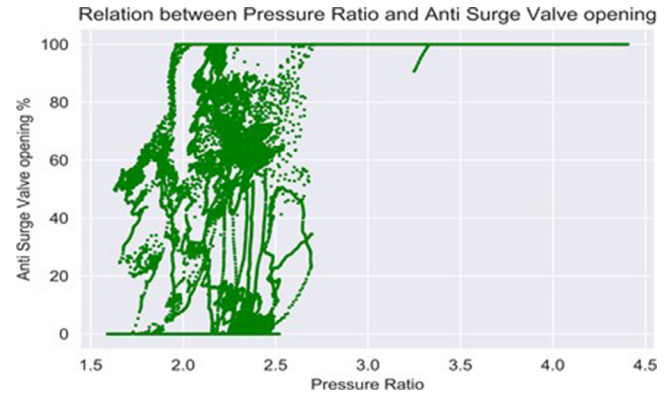


Fig. 6. Paired chart regarding the relation between the pressure ratio and opening of the anti-surge valve.

Table 2
The list of used software and hardware.

Computer characteristics!	"DELL OPTIPLEX 9020 Intel Core i5-4570 CPU 3.2 GHZ Memory 4GB"
"Operating System"	"Windows 7 Ultimate 64 bit"
"Python Version"	3.6.5 Anaconda, Inc
"Keras Version"	2.2.4
"Tensorflow Version"	1.15.0
"Scikit-Learn Version"	0.22.1
"Numpy Version"	1.17.4
"Spyder Version"	4.0.0
"Pandas Version"	0.25.3
"Matplotlib Version"	3.1.1
"Pyod Version"	0.7.7.1

its control system to accomplish this research. Obviously, it can be prepared for any other type of the rotating machine with the same approach.

4.2. Implementation methods and experimental results

In this study, nonlinear regression, novelty detection, outlier detection, and classification algorithms were used to produce the most efficient penetration detection system based on the generated dataset. The performance of these methods was measured and compared based on the response time and different metrics.

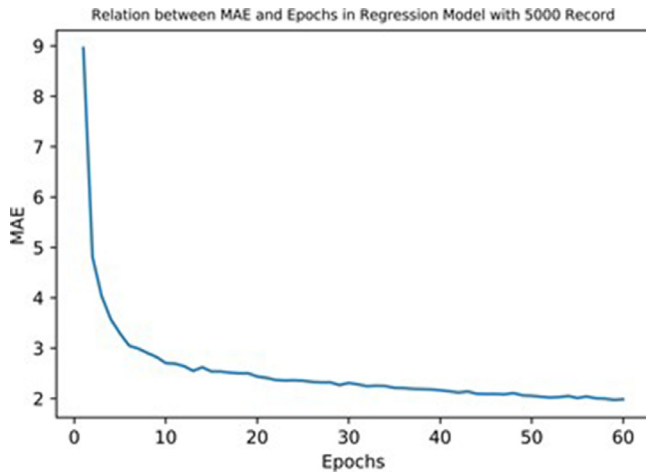
Some hardware and software were used to implement the methods, which are illustrated in Table 2.

In this study, the different famous metrics were used to evaluate and compare the models simultaneously to achieve a real judgment. In the prediction model, the "True Positive (TP)" was used that is defined as the number of positive classified records that are correctly predicted by the model. On the contrary, "False Positive (FP)" is defined as the number of records with a negative

Table 3

Comparison of the nonlinear regression results in different implementation methods.

	No. of Training Record	No. of Test Record	Epochs	Mean Absolute Error	Mean Square Error	Time to Train (s)	Time to Predict (s)
5 Fold Validation	5000	2(00	0	2.00114	29.00~284	215.G29G	0.1648
5 Fold Validation	180,000	2(00	30	1.125	1 G.59G4	3007.3435	0.1887

**Fig. 7.** The relationship between the MAE and Epochs.

class predicted by the model as a positive class. Also, the number of positive records erroneously predicted to be negative is called the "False Negative (FN)" and in contrast, the number of negative records correctly predicted to be negative is called "True Negative (TN)". Given the above definitions, metrics are calculated by the following formulas.

$$\text{Accuracy} = (TP + TN) / (P + N) \quad \text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN) \quad \text{F1} = 2TP / (2TP + FP + FN)$$

Cohen's kappa is a statistic measuring the level of agreement between two evaluators on a classification problem. Cohen's kappa is calculated by the following formula if P_o is the ratio of the observed agreement or the experimental probability of agreement on the label assigned to each sample and also P_e is the expected agreement when both evaluators randomly determine the labels.

$$K = (P_o - P_e) / (1 - P_e)$$

"Area under Curve (AUC) of Receiver Operating Characteristic (ROC)" was also used as another metric.

4.2.1. Nonlinear regression

In the nonlinear regression method, a continuous analog value is predicted based on the inputs. In this case, the goal is to create a model that can predict the opening of the first-stage anti-surge control valve as an analog value based on the other features as input with an acceptable error.

The predicted value can be compared to the actual value to recognize whether the data of the control system have been manipulated by an external agent or intruder or not.

In this section, the best result of nonlinear regression prediction is obtained from the model output, which is a four-layer fully-connected (dense) neural network with the lowest MAE value. In the first case, 180,000 records of the dataset were used to train this neural network model; the best result was achieved when the number of Epochs was below 30 more than which was in over-fitting mode. The number of records was reduced to 5000 to diminish the training time, and it was observed that the best number of Epochs was below 60 as illustrated in Fig. 7.

The 5-fold validation was employed in all the modes to prevent the accidental results. Table 3 compares the cases with 180,000 and 5000 training records where the test dataset is considered with 2000 records.

Considering the most observed error, this method was acceptable to determine the manipulation of the output data in both cases. It should be noted that if the attacker manipulates the first-stage anti-surge control valve aimed at sabotaging the machine, it needs to change the output much more than the error of this model.

4.2.2. Novelty detection

In the novelty detection approach, the "Local Outlier Factor (LOF) and One-Class SVM" algorithms were used to detect the unseen records (here is the abnormal record). The training data set with 180,000 normal records was employed to train these two algorithms. The test dataset with 2000 seen and unseen records was exploited to test the model. The LOF algorithm, as the unsupervised method is used for anomaly and novelty detection that calculates the local density deviation of a given point according to its neighbors. Samples with a lower density than its neighbors are considered abnormal.

The One-Class SVM algorithm has been developed to be used in the unbalanced data sets and can be employed for the cases where it is impossible or difficult to collect the abnormal data. This algorithm was adapted from the SVM algorithm to be utilized with unlabeled data. Table 4 shows the results obtained from the implementation of LOF algorithm with 20 neighbors and One-Class SVM algorithm.

The results of these two algorithms indicated that the One-Class SVM algorithm is more efficient in detecting the novelty, although they were almost identical in terms of time.

4.2.3. Outlier detection

Outlier detection approaches attempt to identify the data that are significantly different from the others. In this study, a number of outlier detection algorithms were utilized, although it is beyond the scope of this study to delve into the details of these methods. Training data set with 180,000 normal records was employed to train each of the algorithms. The records were shuffled in each round of execution to prevent accidental results. The outlier detection algorithms were evaluated by the test dataset with 2000 records created by the random samples from the original dataset, including 180,000 normal records and 20,000 abnormal records. The anomaly ratio to normal data was considered to be 0.01 for algorithms. In this section, each of the outlier detection algorithms "(Angle-Based Outlier Detection (ABOD)-Isolation Forest, Clustering-Based Local Outlier Factor (CBLOF), feature bagging, Histogram-Based Outlier Detection (HBOS), kNN, One-Class SVM, and elliptic envelope)" was implemented in five rounds to prevent random results. Table 5 illustrates the results.

Obviously, the results of the kNN algorithm ranked the first followed by the ABOD in the second rank.

Beside the different metrics the learning curve is used to show the performance over dataset size. Fig. 8 shows the learning curve for Angle-Based Outlier Detection with 2000 sample records.

Table 4

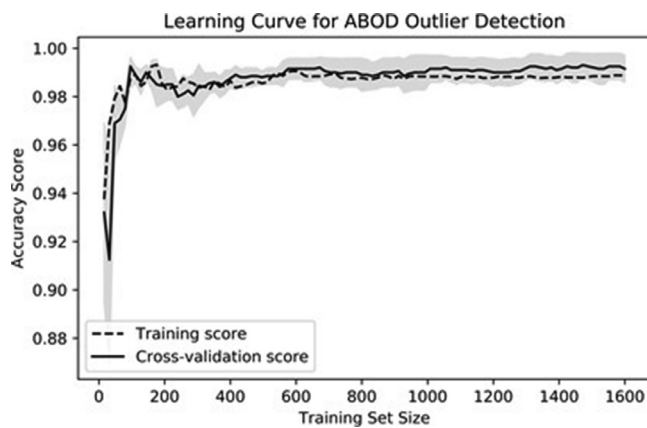
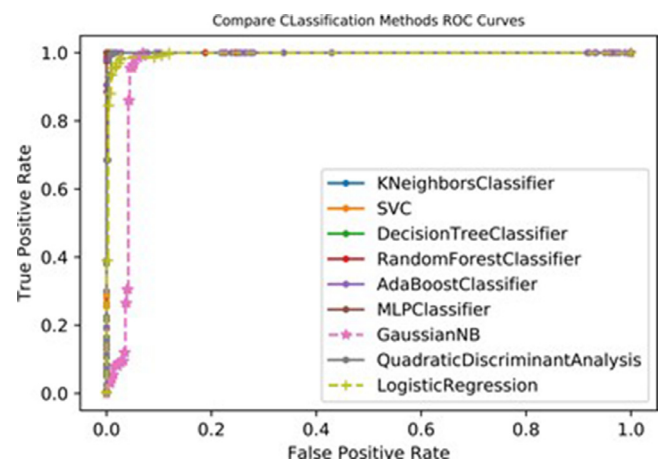
Results of novelty detection methods using the LOF and one-class SVM algorithms.

Method	Accuracy	Precision	Recal	F1 Score	Cohens kappa	ROC AUC	Training Time (s)	Test Time (s)
Local Outlier Factor	0.9547802	0.9387624	0.98004	0.9681078	0.89046 12	0.9380478	42.38072	0.0741)8
One class SVM	0.985764	1	0.9796408	0.9897152	0.966ro62	0.989820'1	52.1851	0.08122

Table 5

Results of outlier detection methods.

Method	Accuracy	Precision	Recal	F1 Score	Cohens kappa	ROC AUC	Training Time (s)	Test Time (s)
ABOD	0.91101	0.9310512	0.11,993,788	0.9638614	0.95813	0.99404 12	57.94154	0.5925
Iforest	0.69982	0.3239314	0.0311784	0.0568732	0.033	0.5102922	16.2352	0.100952
CBLOP	0.8744	0.7079 128	0.1445004	0.23119,674	0.202888	0.567497	8.67652	0
FeatureBagging	0.9847	0.9387572	0.9480666	0.9431802	0.9343408	0.9692456	356.98784	4.39282
HBOS	0.8693	0.3931208	0.0393792	0.071463	0.0490'2	0.5150334	2.3929	0
KNN	0.9945	0.9611558	0.99927	0.97983'26	0.9766484	0.9965176	26.19654	0.65294
one class SVt	0.894208	0.9131382	0.9379244	0.9253646	0.7438736	0.8652498	292.954 18	0.05976
EllipticEnvelope	0.752268	0.764082	0.934 13'2	0.840593	0.309178	0.631799	62.93602	0

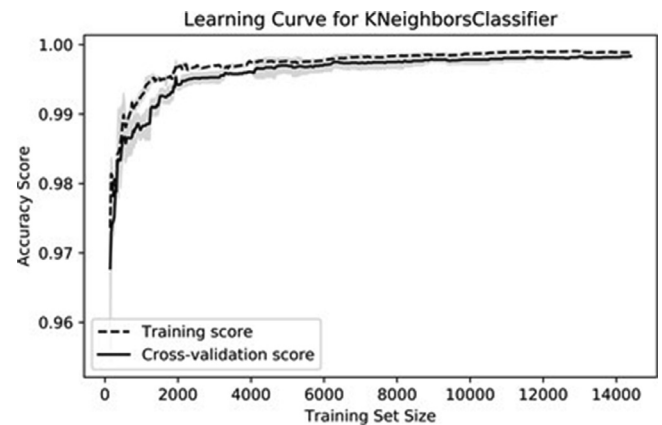
**Fig. 8.** The learning curve for Angle-Based Outlier Detection.**Fig. 9.** The ROC diagram of the classification methods.

4.2.4. Classification

In the classification approaches, the homogeneous records are tried to be placed on the same category, and each record is labeled by the specific class. In this section, a data set containing 180,000 normal records and 29,000 abnormal records was used; consequently, each record was labeled with normal or abnormal class. Since, the number of classes in this data set was not the same; hence, it was necessary to balance the dataset. The dataset was balanced by the less class increasing strategy. The final dataset has consisted of the equivalent class, including, 280,000 records. The test dataset was selected with 2000 records by random sampling from the original dataset. Similar to the previous approaches, the test data set had about 2000 records; therefore, the response time can be comparable in the same conditions for all the approaches. In this section, each of the known classification algorithms "(k-Neighbors classifier, SVC, decision tree classifier, random forest classifier, AdaBoost classifier, MLP classifier, Gaussian Naive Bayes (NB), quadratic discriminant analysis, and logistic regression)" was implemented in five rounds to prevent random results. Table 6 shows the comparison results.

According to the comparison of the results, the decision tree classifier and k-Neighbors classifier methods were in the first rank, followed by the random forest classifier and AdaBoost classifier methods. The comparison was based on the evaluation metrics, training, and test time. Fig. 9 shows the ROC diagram of the classification methods.

The learning curve is employed to illustrate the performance over dataset size. Fig. 10 shows the learning curve for k-Neighbors classifier.

**Fig. 10.** The learning curve for k-Neighbors classifier method.

The following precautions have been taken into account to prohibit creating overfitted model in outlier detection and classification algorithms:

- 5-fold validation is used during training the models and the scores are average of the results. The test data is out of training and validation data.
- The validation curves are utilized to make sure about the method's appropriate parameters.
- The dataset was balanced by the less class increasing strategy before training in classification methods.

Table 6

Results of classification detection methods.

Method	Accuracy	Precision	Roca!	FI Score	Cohens kappa	ROC AUC	Training Time (s)	Test Time (s)
KNeighborsClassifier	0.9999108	0.9007188	1	0.0098592	0.99979'	0.9999348	3.45708	
SVC	0.9999108	1	0.999733	0.009SGG4	0.9007994	0.9998664	1217.38'1'1	0.1GSt G
DecisionTreeClassifier	0.9999108	1	0.9997378	0.9998G88	0.9008014	0.99118G9	1.16404	0.003124
RandomForestClassifier	0.9999108	1	0.90073iG	0.9998GSG	0.9998012	0.9998G88	1.89778	0.0095G
AdaBoostClassifier	0.9999108	0.9997344	1	0.9998672	0.9998	0.999933	19.3'2.52	0.018%7
MLPClassifier	0.9971478	0.99'4008	0.9970824	0.0057384	0.9935942	0.9971368	28.6271	0.007'248
GaussianNB	0.9418734	0.888373	0. 99iG	0.915i2GG	0.8714152	0.942G392	0.1100476	0
QuadraticDiscriminantAnalysis	0.98G270G	0.9615088	0.9%'9438	0.979SG58	0.00!U54G	0.9894264	0.5ro71	0.0125
LogisticRegression	0.95G31G	0.91032G4	0.9631854	0.9359i2G	0.902SG28	0.9580644	3.11928	0.000>

Table 7

Comparison of the results regarding the best detection method among four approaches.

Method	Algorithm	Accuracy	Precision	Recal	FI Score	Cohens kappa	ROC AUC	Training Time	Test Time (s)
Novelty detection	one class SVM	0.985	1	0.979	0.989	0.966	0.989	52.185	0.081
outlier detection	KNN	0.991	0.961	0.999	0.979	0.976	0.996	26.196	0.652
outlier detection	ABOD	0.99	0.931	0.999	0.963	0.958	0.994	57.94 1	0.59'2
classification	KNeighborsClassifier	0.999	0.999	1	0.999	0.999	0.999	3.457	0.308
classification	SVC	0.999	1	0.999	0.999	0.999	0.999	1217.382	0.168
classification	decision tree classifier	0.999	1	0.999	0.999	0.999	0.999	1.164	0
classification	random forest. classifier	0.999	1	0.999	0.999	0.999	0.999	7.897	0.009
classification	AdaBoost classifier	0.999	0.999	1	0.999	0.9998	0.999	19.325	0.ot8
nonlinear regression	mean absolute error 1.5731 mean square error 17.4166							4944.316	0.1882

- The models were trained and validated with different size of training dataset and test them with test records out of training dataset.

Table 7 illustrates the results regarding the best algorithm in the mentioned approach in one scheme for better evaluation.

5. Conclusion

In this paper, several major attacks on ICSs were identified, and the IDS methods developed to overcome the threats of ICS were reviewed. It is unlikely to detect the advanced cyberattacks based on data tampering using the conventional IDS methods as the attacker accomplishes manipulate the system with normal network traffic. Here in, a new method was proposed based on the process behavior regarding one of the most widely used critical machines in the industries. The trajectory evolution of the created dataset was proposed, including the critical output signals and their related data to be utilized in model implementation. The behavioral model of the rotating machine was implemented by the nonlinear regression, novelty detection, outlier detection, and classification algorithm. The nonlinear regression model implemented by the four-layer fully-connected (dense) neural network had the MAE value equal to 1.5%. The accuracy of behavioral models created by the One-Class SVM, novelty detection, kNN outlier detection, Decision Tree Classifier, k- Neighbors Classifier, Random Forest Classifier, and AdaBoost Classifier were obtained as 0.98, 0.994, 0.999, 0.999, 0.999, and 0.999, respectively. The results indicated that the classification and outlier detection are more efficient than the novelty and regression according to other evaluation metrics as well. Test time values demonstrated that all the methods work in a timely manner because of the control system scan time.

Here in, our original dataset only consisting of normal records was used by the nonlinear regression and novelty detection algorithms, which is considered as strength and an advantage for these methods. Test time values revealed that all the methods can work in a timely manner, because their processing time is much less than the scan time of the control system.

The experimental results showed that our proposed method is useful to detect harmful data tampering on the rotating machines. Using the real process data of the rotating machine instead of sim-

ulation or data from the testbed, independency of the control system, and convenient implementation were among the advantages and contributions of this study. The difficulty in selection of the critical output signals and dependency on the experts to choose other related data were the drawbacks of this proposed method.

Declaration of Competing Interest

The authors declare that they have no known competing financial interest or personal relationships that could have appeared to influence that work reported in this paper.

References

- [1] F. Sicard, E. Zamai, J.-M. Flaus, An approach based on behavioral models and critical states distance notion for improving cybersecurity of industrial control systems, *Reliab. Eng. Syst. Saf.* (2019), doi:10.1016/j.ress.2019.03.020.
- [2] S. Han, M. Xie, Y. Ling, Intrusion detection in cyber-physical systems: techniques and challenges, *IEEE Syst. J.* 8 (2014) 1052–1062, doi:10.1109/JSYST.2013.2257594.
- [3] G. Settanni, F. Skopik, A. Karaj, M. Wurzenberger, R. Fiedler, *Protecting Cyber physical production Systems Using Anomaly Detection to Enable Self-Adaptation*, IEEE Industrial Cyber-Physical Systems (ICPS) St. Petersburg, 2018, pp. 173–180.
- [4] I. Friedberg, K. McLaughlin, P. Smith, D. Laverty, S. Sezer, Stpa safesec: safety and security analysis for cyber physical systems, *J. Inf. Secur. Appl.* 34 (2019) 183–196, doi:10.1016/j.jisa.2016.05.008.
- [5] A. Robles-Durazo, Moradpoor N., J. McWhinnie, G. Russell, I. Maneru-Marin, Plc memory attack detection and response in a clean water supply system, *Int. J. Crit. Infrastruct. Prot.* 26 (2019), doi:10.1016/j.ijcip.2019.05.003.
- [6] Threat landscape for industrial automation systems. h1 2019, Tech. rep., Kaspersky Lab ICS-CERT (sep 2019).
- [7] Guide to Industrial Control System (ICS) Security (nist sp 800-82 Rev 2), National Institute of Standards and Technology, 2015 Report, doi:10.6028/NIST.SP.800-82r2.
- [8] J.C. Edward, A. Kott, *Cyber-security of SCADA and Other Industrial Control Systems*, Springer, 2016.
- [9] N. Falliere, L. Murchu, E. Chien, W32.stuxnet dossier, White Paper, Symantec Corporation, 2011.
- [10] X. Clotet, J. Moyano, A. Leon, A real-time anomaly-based ids for cyber-attack detection at the industrial process level of critical infrastructures, *Int. J. Crit. Infrastruct. Prot.* 23 (2018) 11–20, doi:10.1016/j.ijcip.2018.08.002.
- [11] S. Adepu, A. Mathur, Using process invariants to detect cyber attacks on a water treatment system, in: In: Hoepman J.H. Katzenbeisser S. (eds) *ICT Systems Security and Privacy Protection*. SEC 2016. IFIP Advances in Information and Communication Technology, Vol. 471, Springer, Cham, 2016, pp. 91–104. doi:10.1007/978-3-319-33630-5_7.
- [12] D. Ding, Q. Han, Y. Xiang, X. Ge, X. Zhang, A survey on security control and attack detection for industrial cyber-physical systems, *Neurocomputing* 275 (2017) 1674–1683, doi:10.1016/j.neucom.2017.10.009.

- [13] P. Cheng, H. Zhang, J. Chen, *Cyber Security for Industrial Control Systems from the Viewpoint of Close-Loop*, CRC Press Taylor & Francis Group, 2016.
- [14] H. Li, J. Zhang, X. He, Design of data-injection attacks for cyber physical systems based on kullback leibler divergence, *Neurocomputing* 361 (2019) 77–84, doi:[10.1016/j.neucom.2019.05.085](https://doi.org/10.1016/j.neucom.2019.05.085).
- [15] S. Magdi, M. Mutaz, A.B. Uthman, Modeling and control of cyber-physical systems subject to cyber-attacks: a survey of recent advances and challenges, *Neurocomputing* 338 (2019) 101–115, doi:[10.1016/j.neucom.2019.01.099](https://doi.org/10.1016/j.neucom.2019.01.099).
- [16] S. Milinkovic, L. Lazi, Industrial plc security issues, in: 2012 20th Telecommunications Forum (TELFOR), Belgrade, 2012, pp. 1536–1539, doi:[10.1109/TELFOR.2012.6419513](https://doi.org/10.1109/TELFOR.2012.6419513).
- [17] Z. Basnight, J. Butts, J. Lopez, T. Dube, Firmware modification attacks on programmable logic controllers, *Int. J. Crit. Infrastruct. Prot.* 6 (2013) 76–84, doi:[10.1016/j.ijcip.2013.04.004](https://doi.org/10.1016/j.ijcip.2013.04.004).
- [18] C. Schuett, J. Butts, S. Dunlap, An evaluation of modification attacks on programmable logic controllers, *Int. J. Crit. Infrastruct. Prot.* 7 (2014) 61–68, doi:[10.1016/j.ijcip.2014.01.004](https://doi.org/10.1016/j.ijcip.2014.01.004).
- [19] C. Tsang, S. Kwong, Multi-agent intrusion detection system in industrial network using ant colony clustering approach and unsupervised feature extraction, in: 2005 IEEE International Conference on Industrial Technology, 2005.
- [20] S. Cheung, B. Dutertre, M. Fong, U. Lindqvist, K. Skinner, A. Valdes, Using model-based intrusion detection for scada networks, in: *Proceedings of the SCADA Security Scientific Symposium*, 46, 2007, pp. 1–12.
- [21] P. Oman, M. Phillips, Intrusion detection and event monitoring in scada networks, in: E. Goetz, S. Shenoi (Eds.), *Critical Infrastructure Protection*, Springer US, Boston, MA, 2008, pp. 161–173.
- [22] I. Fovino, A. Carcano, T. Murel, A. Trombetta, M. Masera, Modbus/dnp3 state-based intrusion detection system, in: 2010 24th IEEE International Conference on Advanced Information Networking and Applications, 2010, pp. 729–736, doi:[10.1109/AINA.2010.86](https://doi.org/10.1109/AINA.2010.86).
- [23] Y. Yang, T. Littler, H. Wang, K. McLaughlin, S. Sezer, Rule-based intrusion detection system for scada networks, in: 2013 2nd IET Renewable Power Generation Conference, 2013, pp. 729–736, doi:[10.1049/cp.2013.1729](https://doi.org/10.1049/cp.2013.1729).
- [24] W. Gao, T. Morris, B. Reaves, D. Richey, On scada control system command and response injection and intrusion detection, in: 2010 eCrime Researchers Summit, 2010, pp. 1–9, doi:[10.1109/ecrime.2010.5706699](https://doi.org/10.1109/ecrime.2010.5706699).
- [25] M. Wu, Z. Song, Y. Moon, detecting cyber-physical attacks in cyber manufacturing systems with machine learning methods, *J. Intell. Manuf.* 30 (2019) 1111–1123, doi:[10.1007/s10845-017-1315-5](https://doi.org/10.1007/s10845-017-1315-5).
- [26] D. Ding, Q. Han, Z. Wang, X. Ge, A survey on model-based distributed control and filtering for industrial cyber physical systems, *IEEE Trans. Ind. Inform.* 15 (5) (2019) 2483–2499, doi:[10.1109/TII.2019.2905295](https://doi.org/10.1109/TII.2019.2905295).
- [27] N. Moustafa, J. Hu, J. Slay, A holistic review of network anomaly detection systems: a comprehensive survey, *J. Netw. Comput. Appl.* 128 (2019) 33–55, doi:[10.1016/j.jnca.2018.12.006](https://doi.org/10.1016/j.jnca.2018.12.006).
- [28] D. Urbina, J. Giraldo, A. Cardenas, J. Valente, M. Faisal, Survey and New Directions for Physics-Based Attack Detection in Control Systems, National Institute of Standards and Technology, 2016 Report, doi:[10.6028/NIST.GCR.16-010](https://doi.org/10.6028/NIST.GCR.16-010).
- [29] S. Zhanwei, L. Zenghui, Abnormal detection method of industrial control system based on behavior model, *Comput. Secur.* 84 (2019) 166–178, doi:[10.1016/j.cose.2019.03.009](https://doi.org/10.1016/j.cose.2019.03.009).
- [30] F. Zhang, H.A.D.E. Kodituwakku, J.W. Hines, J. Coble, Multilayer data-driven cyber-attack detection system for industrial control systems based on network, system, and process data, *IEEE Trans. Ind. Inform.* 15 (7) (2019) 4362–4369, doi:[10.1109/TII.2019.2891261](https://doi.org/10.1109/TII.2019.2891261).
- [31] M. Kravchik, A. Shabtai, Detecting cyber attacks in industrial control systems using convolutional neural networks, in: *Proceedings of the 2018 Workshop on Cyber-Physical Systems Security and Privacy*, 2018, pp. 72–83, doi:[10.1145/3264888.3264896](https://doi.org/10.1145/3264888.3264896).
- [32] J. Li, X. Huang, Cyber attack detection of i&c systems in npps based on physical process data, in: *Proceedings of the 2016 24th International Conference on Nuclear Engineering ICONE24*, 2016, doi:[10.1115/ICONE24-60773](https://doi.org/10.1115/ICONE24-60773).
- [33] S. Adepu, A. Mathur, Distributed detection of single-stage multipoint cyber attacks in a water treatment plant, in: *Proceedings of the 11th ACM on Asia Conference on Computer and Communications Security*, 2016, pp. 449–460, doi:[10.1145/2897845.2897855](https://doi.org/10.1145/2897845.2897855).
- [34] M. Wu, Y.B. Moon, Intrusion detection system for cyber-manufacturing system, *J. Manuf. Sci. Eng.* 141 (3) (2019), doi:[10.1115/1.4042053](https://doi.org/10.1115/1.4042053).
- [35] X. Huang, D. Zhai, J. Dong, Adaptive integral sliding-mode control strategy of data-driven cyber-physical systems against a class of actuator attacks, *IET Control Theory Appl.* 12 (2018) 1440–1447, doi:[10.1049/iet-cta.2017.1278](https://doi.org/10.1049/iet-cta.2017.1278).
- [36] D. Hadziosmanovic, R. Sommer, E. Zamboni, P. Hartel, Through the eye of the plc: semantic security monitoring for industrial processes, in: *Proceedings of the 30th Annual Computer Security Applications Conference on ACSAC*, 14, 2014, pp. 126–135, doi:[10.1145/2664243.2664277](https://doi.org/10.1145/2664243.2664277).
- [37] W. Aoudi, M. Iturbe, M. Almgren, Truth will out: departure-based process-level detection of stealthy attacks on control systems, in: *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security (CCS '18)*, Association for Computing Machinery, 2018, pp. 817–831, doi:[10.1145/3243734.3243781](https://doi.org/10.1145/3243734.3243781).
- [38] D. Urbina, G. Giraldo, A. Cardenas, N. Tippenhauer, J. Valente, M. Faisal, J. Ruths, R. Candell, Limiting the impact of stealthy attacks on industrial control systems, in: *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 2016, pp. 1092–1105, doi:[10.1145/2976749.2978388](https://doi.org/10.1145/2976749.2978388).
- [39] A. A. Cardenas, A. Saurabh, Z.-S. Lin, Y.-L. Huang, C.-Y. Huang, S. Sastry, Attacks against process control systems: risk assessment, detection, and response, in: *Proceedings of the 6th ACM Symposium on Information, Computer and Communications Security*, 2011, pp. 355–366, doi:[10.1145/1966913.1966959](https://doi.org/10.1145/1966913.1966959).
- [40] G. McMillan, *Centrifugal and Axial Compressor Control*, Momentum Press, 2010.